Determination of the g-factor for an electron using Electron Spin Resonance Spectrometer

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1 Abstract

Electron Spin Resonance (ESR) spectroscopy, also known as Electron Paramagnetic Resonance (EPR) spectroscopy, is a powerful technique used to study the electronic structure and properties of paramagnetic species. One of the fundamental applications of ESR spectroscopy is the determination of the g-factor, which provides critical insights into the magnetic behavior of free electrons. This report contains the principles behind ESR spectroscopy and its application in the determination of the g-factor for free electrons. We will discuss the experimental setup, data analysis.

2 Apparatus used

- Electron Spin Resonance apparatus (12MHz–16MHz R.F. Oscillator and Phase shifter)
- Digital Cathode Ray Oscilloscope
- Helmholtz coil and sample holder with R.F. coil
- DPPH sample
- Connecting cables

3 Theory & Experimental Set-up

The g-factor, often referred to as the electron g-factor (gyromagnetic factor), is a dimensionless constant that characterizes the relationship between an electron's magnetic moment and its angular momentum. The determination of the g-factor for a free electron is of paramount importance in quantum physics, as it serves as a validation of theoretical predictions and provides a foundation for understanding electron behavior in various contexts. In simple classical concepts, we can understand the fundamentals of this technique.

Let's suppose a particle having magnetic moment $\vec{\mu}$ is placed in a uniform magnetic field of intensity $\vec{H_0}$ (it is in the direction of the z-axis), and the moment is processing around the magnetic field with an angular Larmor frequency given by,

$$\omega_0 = g\left(\frac{e}{2mc}\right)H_0 \tag{1}$$

Here, g is the g-factor ($g = 1$ for pure orbital angular momentum and $g = 2$ for a free electron spin). This effective g factor is known as the spectroscopic splitting factor. Now, that system is given an additional weak magnetic field $\vec{H_1}$ oriented in the xy plane and rotating about the Z axis with an angular frequency ω_1.

If ω_0 is different from ω_1, then the angle between the field $\vec{H_1}$ and the magnetic moment $\vec{\mu}$ will continuously change so their interaction will average out to zero. But if $\omega_0 = \omega_1$, then the angle between $\vec{H_1}$ and $\vec{\mu}$ is maintained, and the net interaction is not zero.

So, we can say that the change in angle corresponds to the change in the potential energy of the particle in the magnetic field (this is a classical analogy to a transition). We will see that the transitions will happen when the rotating field has an angular frequency $\omega_0 = \omega_1$.

Now in the quantum domain, the electron has an intrinsic spin angular momentum $\vec{S}$ which couples with the orbital angular momentum of electron $\vec{L}$ and gives the resultant $\vec{J}$, and there are $J + 1$ magnetic sublevels labeled by the field $\vec{H_0}$ with equal energy difference,

$$\Delta E = g_L \mu_0 H_0 \tag{2}$$

Here, μ_0 is Bohr magneton, and g_L is the Lande g-factor. The quantum mechanical value of g_L is given by,

$$g_L = 1 + \frac{J(J+1) + S(S+1) - L(L+1)}{2J(J+1)} \tag{3}$$

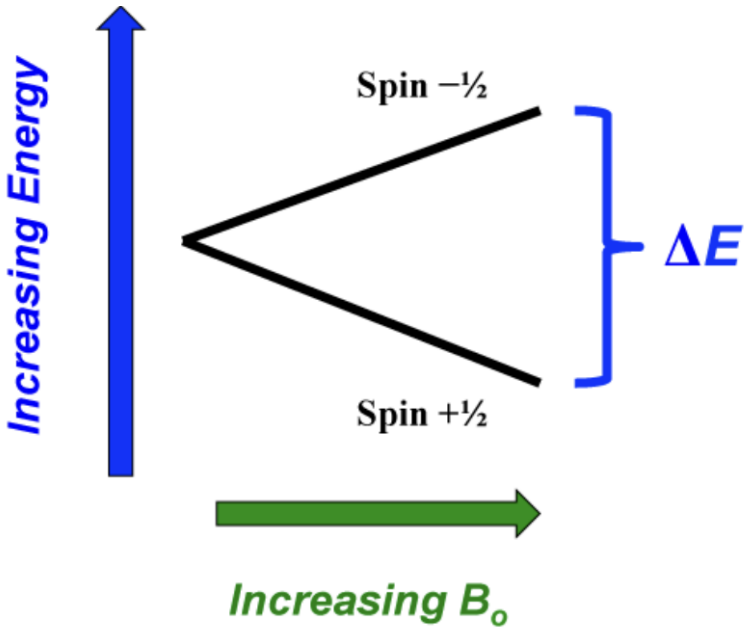


Figure 1: Zeeman Splitting

Now, if the particle is subjected to a perturbation by an alternating magnetic field with a frequency ν, such that the quantum $h\nu$ is exactly the same as the difference between the levels, ΔE, and if the direction of the alternating field is perpendicular to the direction of the static magnetic field, then there will be induced transitions between neighboring sub-levels according to the selection rules $\Delta m = \pm 1$ for magnetic dipolar radiation. Therefore, the condition of resonance is

$$\Delta E = g\mu_0 H_0 = h\nu_1 \tag{4}$$

Here, ν_1 is the resonant frequency. In atomic spectroscopy, we can't observe the transitions between sub-levels with different m, but we can observe the splitting of levels through a small change in the frequency of the radiation emitted in the transition between widely distant levels.

The magnetic field at the center of a Helmholtz coil is given by:

$$H = \frac{32\pi n}{10\sqrt{125}a} I \tag{5}$$

Now, the resonance magnetic field that is causing the spin resonance is given by:

$$H_0 = \frac{Q}{P \cdot 2\sqrt{2}H} \tag{6}$$

Here, P and Q are parameters obtained from the 4-peak resonance curve in the CRO display screen. We know that if resonance is happening at frequency ν_0, then:

$$h\nu_0 = g\mu_B H_0 \tag{7}$$

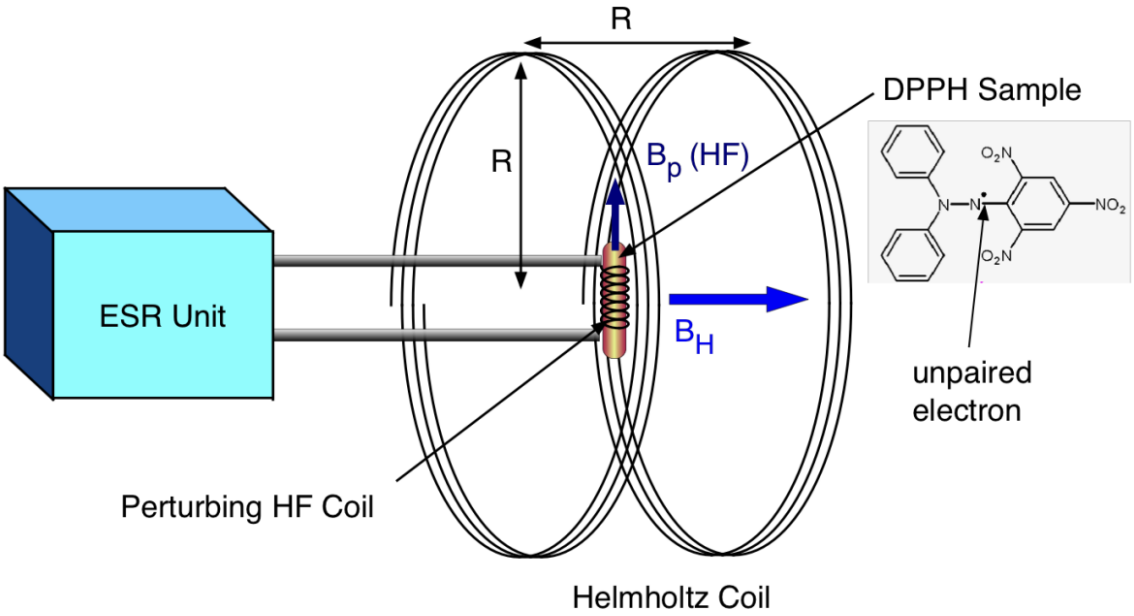


Figure 2: Experimental Set-up

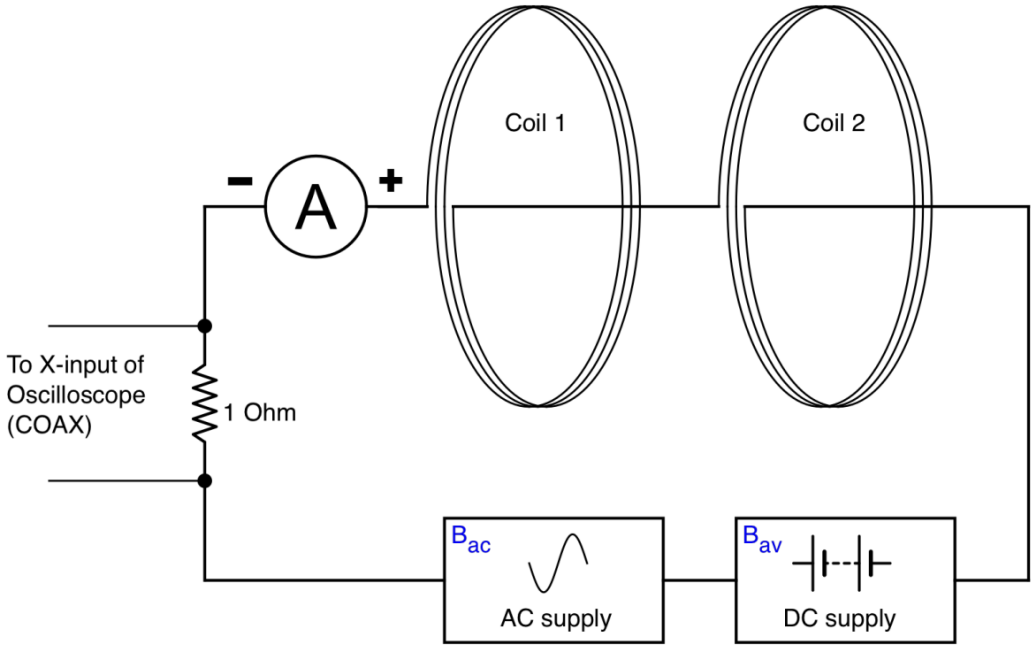


Figure 3: Circuit diagram of the Helmholtz coil

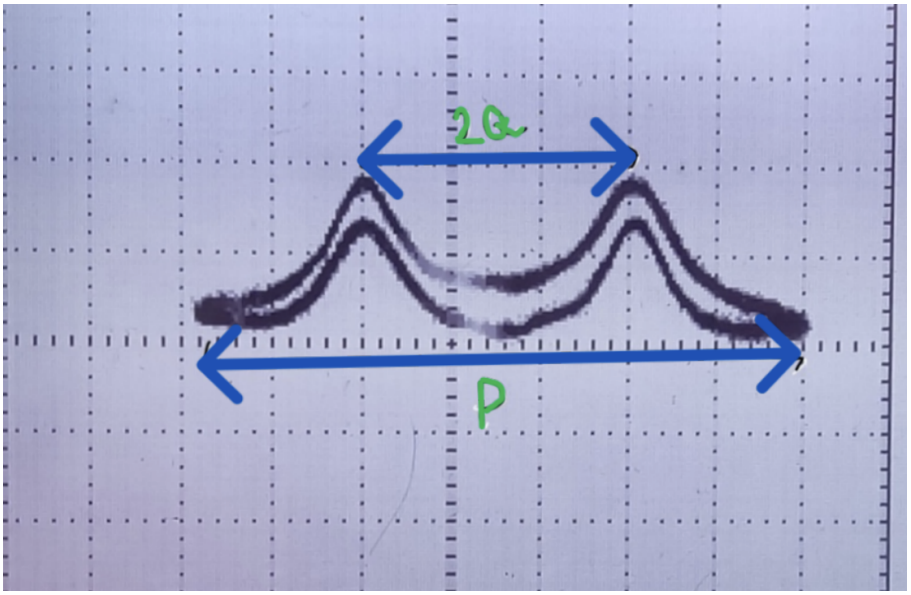


Figure 4: CRO display

4 Observations & Data Analysis

where g is the Lande g-factor (here there is only a spin contribution), and μ_B is the Bohr magneton. Thus, we have:

$$g = \frac{h\nu_0}{\mu_B} \cdot \frac{P}{Q} \cdot \frac{1}{2\sqrt{2}H} \tag{8}$$

Now, in our case, we have $n = 500$ and $a = 7.7$ cm, and hence:

$$\frac{H}{I} = \frac{32 \times \pi \times 500}{10\sqrt{125 \times 7.7}}\text{Gauss/Amp} = 58.388\text{Gauss/Amp} \tag{9}$$

Hence, we have:

$$g = \frac{h\nu_0}{\mu_B} \cdot \frac{P}{IQ} \cdot \frac{1}{165.147} \tag{10}$$

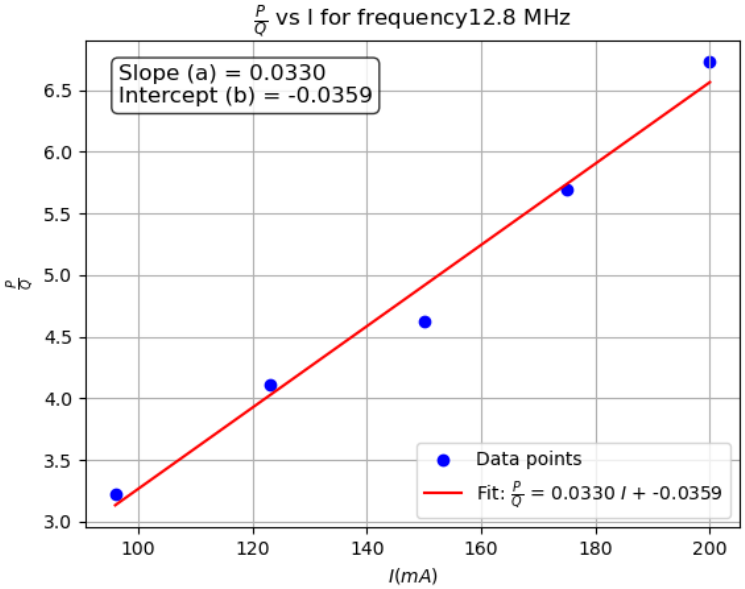
$$= (7.145 \times 10^{-7})\nu_0 \cdot \frac{P}{IQ} \cdot \frac{1}{165.147} = (4.326 \times 10^{-9})\nu_0 \cdot \frac{P}{IQ} \tag{11}$$

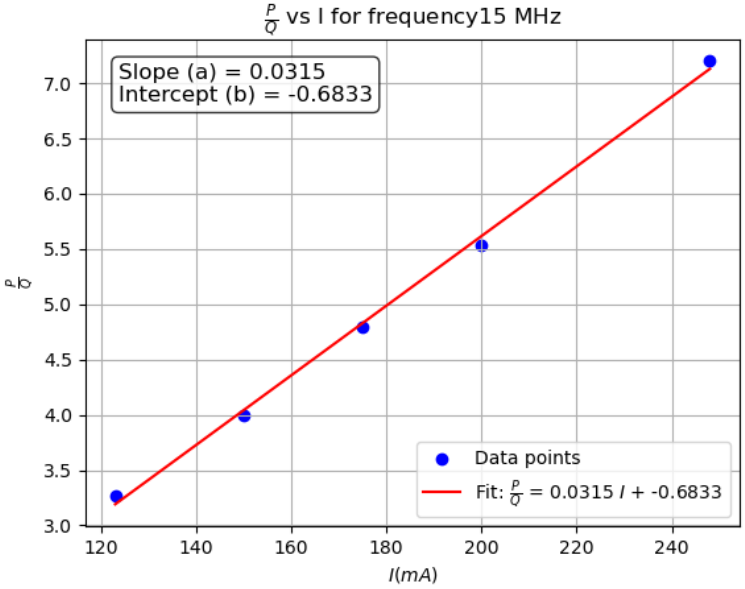
For each value of ν_0, we can perform the following steps:

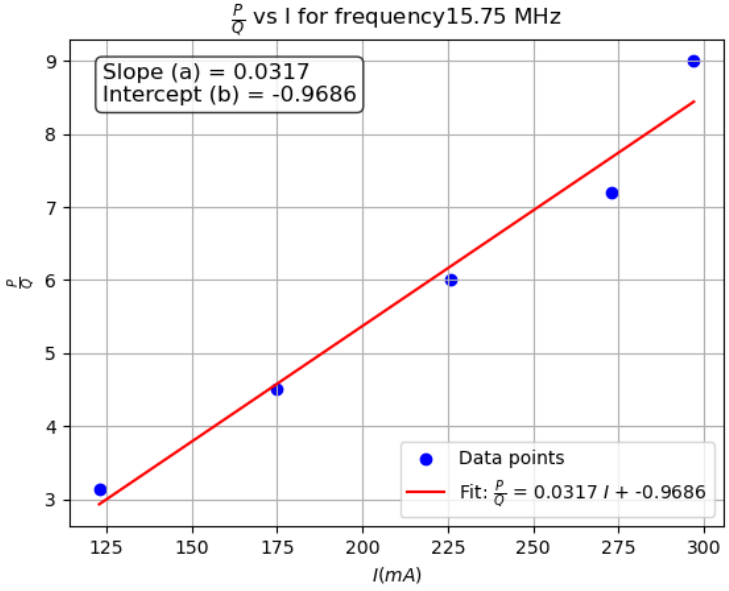
1. Creating a plot where the x-axis represents the current I and the y-axis represents P/Q. Each point on the plot corresponds to a specific resonant frequency ν_0.
2. Fitting a straight line to the plotted data points. The slope of this fitted line provides us with the value of $P/(I \cdot Q)$.
3. Calculating the g-factor for each resonant frequency ν_0 based on the obtained value of $P/(I \cdot Q)$.
4. Calculating the average of all the g-factors obtained for each resonant frequency value. This average g-factor comprehensively characterizes the system's behavior across different frequencies.

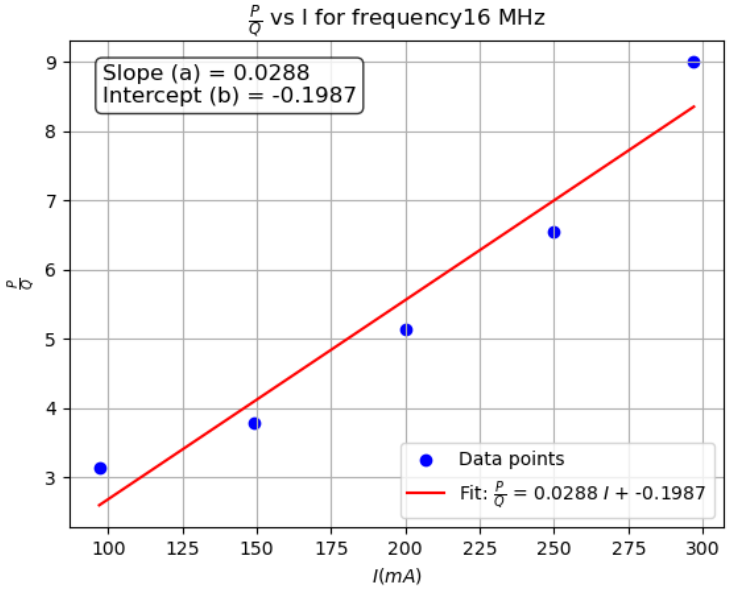
This approach allows us to determine the g-factor with increased accuracy by considering multiple resonant frequency measurements and obtaining an averaged value.

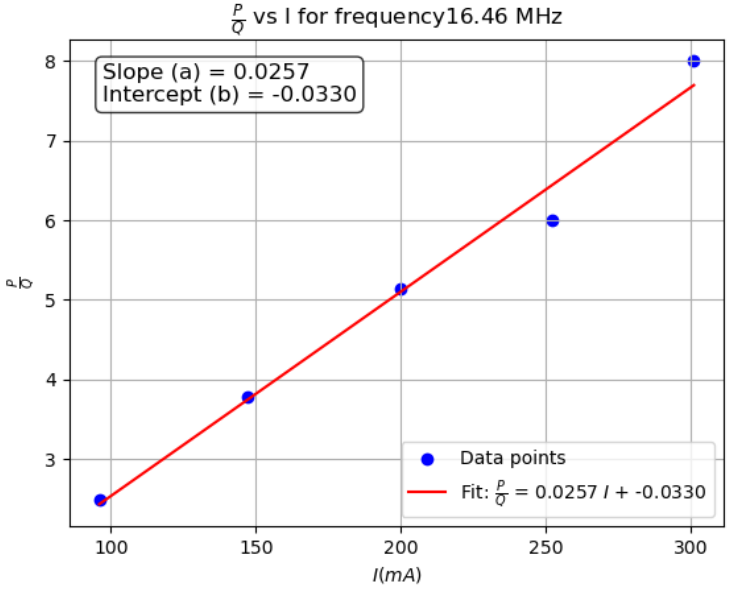
4.1 Plots & fitting






$\frac{P}{Q}$ vs I for frequency15.75 MHz
Slope (a) = 0.0317
Intercept (b) = -0.9686
Data points
Fit: $\frac{P}{Q}$ = 0.0317 I + -0.9686
$\frac{P}{Q}$
I(mA)


$\frac{P}{Q}$ vs I for frequency16 MHz
Slope (a) = 0.0288
Intercept (b) = -0.1987
Data points
Fit: $\frac{P}{Q}$ = 0.0288 I + -0.1987
$\frac{P}{Q}$
I(mA)


$\frac{P}{Q}$ vs I for frequency16.46 MHz
Slope (a) = 0.0257
Intercept (b) = -0.0330
Data points
Fit: $\frac{P}{Q}$ = 0.0257 I + -0.0330
$\frac{P}{Q}$
I(mA)

4.2 Data Analysis

Freq(MHz)	I(mA)	2Q	P	P/Q	Slope[(P/Q) vs I]	G factor
12.8	96	23	37	3.217391	0.033001	1.827998
	123	18	37	4.111111		
	150	16	37	4.625		
	175	13	37	5.692308		
	200	11	37	6.727273		
15	123	22	36	3.272727	0.031504	2.045002
	150	18	36	4		
	175	15	36	4.8		
	200	13	36	5.538462		
	248	10	36	7.2		
15.75	123	23	36	3.130435	0.031694	2.160204
	175	16	36	4.5		
	226	12	36	6		
	273	10	36	7.2		
	297	8	36	9		
16	97	23	36	3.130435	0.028803	1.994336
	149	19	36	3.789474		
	200	14	36	5.142857		
	250	11	36	6.545455		
	297	8	36	9		
16.46	96	29	36	2.482759	0.025683	1.829404
	147	19	36	3.789474		
	200	14	36	5.142857		
	252	12	36	6		
	301	9	36	8		
Average G factor						**1.971389**

5 Error Analysis

Sources of Error:

1. **Instrumental Errors:** Errors related to the precision and accuracy of the measurement instruments used in the experiment can introduce uncertainties in the results.

2. **Fluctuations in Current Values:** Variations or fluctuations in the current values supplied to the Helmholtz coils can lead to inconsistencies in the measurements and affect the reliability of the data.

3. **Insufficient Data Points:** The experiment may suffer from limited data points, and taking fewer current readings might not maximize the accuracy of the mea-

surements. This can result in a less comprehensive understanding of the system's behavior.

4. **Human Error in Recording Readings:** Human errors in noting down readings from the oscilloscope gridlines, such as misinterpretation or misrecording, can introduce inaccuracies in the collected data.

From Eqn. (8):

$$\frac{\delta g}{g} = \frac{\delta \nu_0}{\nu_0} + \frac{\delta P}{P} + \frac{\delta Q}{Q} + \frac{\delta H}{H} \tag{12}$$

Now, for $\frac{\delta H}{H}$, we have:

$$\frac{\delta H}{H} = \frac{\delta a}{a} + \frac{\delta I}{I} \approx \frac{0.1}{7.7} + \frac{1}{100} \approx 0.023 \tag{13}$$

So, we can write:

$$\frac{\delta g}{g} \approx \frac{0.01}{15.202} + \frac{1}{36.2} + \frac{0.5}{7.72} + 0.023 \approx 0.116 = 11.6\% \tag{14}$$

So we can say,

$$\Rightarrow \delta g \approx 1.971389 \times 0.116 \approx 0.22 \tag{15}$$

6 Code for fitting

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.optimize import curve_fit

def linear_func(x, a, b):
    return a * x + b


def plot_and_fit(X, Y,z):
    X = np.array(X)
    Y = np.array(Y)

    # Perform linear curve fitting
    popt, _ = curve_fit(linear_func, X, Y)
    a, b = popt

    # Generate fitted Y values
    Y_fit = linear_func(X, a, b)
    print(a)
    # Plot data points
    plt.scatter(X, Y, color='blue', label='Data points')
    # Plot fitted linear curve
    plt.plot(X, Y_fit, color='red',
             label=f'Fit: $\\frac{{P}}{{Q}}$ = {a:.4f} $I$ + {b:.4f}')
```

```
    # Display fit parameters on the plot
    plt.text(0.05, 0.95, f'Slope (a) = {a:.4f}\nIntercept (b) = {b:.4f}',
             transform=plt.gca().transAxes, fontsize=12, verticalalignment='top',
             bbox=dict(boxstyle='round', facecolor='white', alpha=0.8))

    plt.xlabel(r'$I(mA)$')
    plt.ylabel(r'$\frac{P}{Q}$')
    plt.title(r'$\frac{P}{Q}$ vs I for frequency' + str(z) + ' MHz')
    plt.legend()
    plt.grid(True)
    plt.show()

X=[96, 123, 150, 175, 200, 123, 150, 175, 200, 248, 123, 175,
226, 273, 297, 97, 149, 200, 250, 297, 96, 147, 200, 252, 301]

Y=[3.217391, 4.111111, 4.625, 5.692308, 6.727273, 3.272727, 4,
4.8, 5.538462, 7.2, 3.130435, 4.5, 6, 7.2, 9, 3.130435, 3.789474,
5.142857, 6.545455, 9, 2.482759, 3.789474, 5.142857, 6, 8]

freq=[12.8, 15, 15.75, 16, 16.46]

for i in range(len(freq)):
    plot_and_fit(X[5*i:(5*i+5)],Y[5*i:(5*i+5)],freq[i])
```

7 Results & Precautions

Thus, we report the spin g-factor of an electron to be: 1.97 ± 0.22. While performing the experiment, we should take care of the following:

• During the experiment, it's crucial to ensure that the peaks of the two curves align or superimpose perfectly, indicating that they are in phase. We can achieve this alignment by adjusting the phase knob on the oscilloscope.

• To avoid potential damage to the Helmholtz coils, it's important not to increase the current to excessively high levels, as this can lead to overheating of the coils.

• It's crucial to be mindful of the wires on the table because even a minor disruption can introduce significant noise into the signals, potentially leading to distortion on the oscilloscope. This is especially critical when working with sensitive radio frequency signals.

• X and Y plate sensitivities of the Oscilloscope should be adjusted such that they should be in the linear range only.

• Since the Helmholtz current is unstabilized, care should be taken that it should remain constant during the observation.

8 Discussion

The experimentally determined average g-factor for the electron in the DPPH sample is 1.971 ± 0.22, calculated from linear fits of P/Q versus current I across multiple resonant frequencies (12.8 MHz to 16.46 MHz). This value is close to the theoretical g-factor for a free electron, which is approximately 2.0023, with our result falling within about 1.6% of the expected value when considering the uncertainty.

$$\frac{|2.0023 - 1.971|}{2.0023} \times 100\% \approx 1.6\%.$$

For DPPH (diphenylpicrylhydrazyl), a common organic radical used in ESR experiments, the literature g-factor is typically around 2.0036, indicating a slight underestimation in our measurement.

For data analysis, linear fitting of P/Q versus I was performed using Python with the `scipy.optimize.curve_fit` function to obtain slopes for each frequency dataset. This method minimized least-squares residuals, providing robust estimates of $P/(I \cdot Q)$ used in the g-factor formula (Equation 10). The code implemented a simple linear model ($y = mx + c$, with c ideally near zero), and fit uncertainties were considered negligible compared to measurement errors. An example of the fitting code is provided earlier for reproducibility.

Overall, the results validate the ESR technique for g-factor determination, with deviations attributable to experimental constraints rather than fundamental flaws in the methodology. Future improvements could include using a purer sample or more precise current stabilization to reduce uncertainty.